Multi-Turn Distributed Prompt Injection Detection

Kyriakos “Rock” Lambros

COMP 4531: Deep Learning, Spring 2026

Abstract

Single-turn prompt injection classifiers achieve high accuracy on known attack distributions, yet they fail against multi-turn attacks that distribute malicious intent across several conversation turns, where each turn appears benign in isolation and the adversarial signal emerges only from how turns relate over time. We present a dual-encoder architecture pairing a frozen single-turn GRU encoder (2.6M parameters) with a trainable sequence-level LSTM (27K trainable parameters) that carries accumulated context across turns. On 73,390 single-turn samples from eight public datasets, the Chollet heuristic correctly predicts that a TF-IDF bag-of-bigrams classifier (F1 = 0.834) outperforms all deep learning models, including transformers, at this data scale. On a shared-prefix dataset of 27,180 synthetic multi-turn conversations generated with Claude Sonnet 4.6 across four difficulty tiers, the temporal LSTM reaches F1 = 0.837 (95% bootstrap CI [0.826, 0.847]) and significantly outperforms turn-level max-voting by +0.131 F1 (paired bootstrap, $p < 0.001$) using the same frozen encoder. Shuffling the turn order of correctly classified attacks flips 55% to incorrect, the strongest evidence that the model learns genuine temporal patterns rather than bag-of-turns features. A 66.4M-parameter concatenated DistilBERT reaches F1 = 0.992 with 2,460x more trainable parameters; the contribution here is not absolute accuracy but the demonstration that temporal modeling adds significant, ordering-dependent signal beyond per-turn classification in a 27K-parameter model deployable on resource-constrained devices.

Contents

1. Problem Statement and Motivation

1.1 The Problem

Single-turn prompt injection detection achieves high accuracy on known attack patterns. ProtectAI's DeBERTa model reaches 99%+ F1 on published benchmarks, yet this performance is limited to known attack distributions. Novel attacks, adversarially crafted inputs, and multi-turn strategies routinely evade these detectors (InjecGuard, 2024; Vassilev, 2025). Real-world attacks against AI agent systems increasingly distribute malicious intent across multiple conversation turns, where each individual turn appears benign in isolation.

This is the **Crescendo attack pattern** (Russinovich et al., USENIX Security 2025): an attacker gradually escalates across turns, establishing trust and context before making the exploitative request. The **Foot-in-the-Door technique** (EMNLP 2025) similarly exploits compliance momentum across turns. Vassilev (2025) extends Gödel's incompleteness theorem to argue that no single-turn classifier can theoretically detect all such attacks.

1.2 Why Deep Learning

The multi-turn attack signal is fundamentally temporal: earlier turns create context that later turns exploit. LSTM and GRU architectures are designed to model exactly this type of sequential dependency through their gating mechanisms. A bag-of-words or per-turn classifier has no mechanism to carry forward accumulated risk state.

1.3 Contribution

This project builds a multi-turn distributed prompt injection detection system using a dual-encoder architecture: a frozen single-turn encoder paired with a sequence-level LSTM that carries forward context across turns. We evaluate on a shared-prefix dataset of 27,180 synthetic conversations across four difficulty tiers, with bootstrap confidence intervals and paired significance tests for all comparisons.

---

2. Data and Preprocessing

2.1 Single-Turn Datasets

Eight HuggingFace datasets were merged:

Dataset	Samples	Type
deepset/prompt-injections	662	Binary classification
xTRam1/safe-guard-prompt-injection	10,296	Binary classification
neuralchemy/Prompt-injection-dataset	6,274	Binary classification
imoxto/prompt_injection_cleaned_dataset-v2	40,000 (subsampled from 535K)	Binary with system prompt context
reshabhs/SPML_Chatbot_Prompt_Injection	16,012	Gandalf CTF attacks

Dataset	Samples	Type
TrustAIRLab/in-the-wild-jailbreak-prompts (jailbreak)	1,405	Real-world jailbreaks
TrustAIRLab/in-the-wild-jailbreak-prompts (regular)	13,735	Real-world benign prompts
jackhhao/jailbreak-classification	1,306	Curated binary

After cleaning (deduplication, whitespace normalization, length filtering), 73,390 samples remained, split 70/15/15 stratified on label: 51,373 train / 11,008 val / 11,009 test. Class balance: ~64% benign / 36% injection.

2.2 Cleaning Pipeline

Nine cleaning steps applied in order: column normalization, label normalization (0=benign, 1=injection), whitespace stripping, internal whitespace collapse, exact deduplication, near-deduplication (lowercase + strip punctuation), short text removal (<3 tokens), long text removal (>2048 chars), and logging.

Removals: 167 exact duplicates, 40 near-duplicates, 132 empty/short, 489 too-long.

2.3 Synthetic Multi-Turn Data (v3 Shared-Prefix)

No public dataset of multi-turn distributed attacks exists. We generated **27,180 synthetic conversations** (18,754 train / 3,296 val / 5,130 test) using a shared-prefix architecture and the Anthropic API (Claude Sonnet 4.6).

Shared-prefix design: Each conversation is generated as a matched pair: one benign continuation and one attack continuation branching from an identical conversational prefix. The prefix length k is sampled uniformly from $\{3, 4, 5\}$ user turns. Both continuations run for 3-5 additional turns, producing conversations of 6-9 user turns total (12-19 turns including assistant responses). This paired structure eliminates vocabulary-level confounds in the opening turns: a first-turn-only classifier achieves F1 = 0.35 (chance level).

Attack strategies:

Strategy	% of Attacks	Pattern	Research Basis
Fragment distribution	45%	Split injection across 3-5 turns, interleaved with on-topic filler	Evasion of per-message filters
Gradual escalation	25%	Crescendo pattern, where each turn nudges toward the attack goal	Russinovich et al. (USENIX Security 2025)
Context priming	15%	Establish persona/authority early, exploit later	Foot-in-the-Door (EMNLP 2025)

Strategy	% of Attacks	Pattern	Research Basis
Instruction layering	15%	Each turn adds one constraint, cumulatively overriding safety	Incremental constraint injection

Difficulty tiers: Each attack is assigned a tier (easy, medium, hard, adversarial) controlling how aggressively the injection signal is obscured. All splits are balanced 50/50 within each tier.

Tier	Train	Val	Test
Easy	5,812	1,002	1,462
Medium	5,684	1,000	1,414
Hard	5,590	976	1,394
Adversarial	1,668	318	860

Data quality controls: (1) Shared-prefix pairing eliminates early-turn confounds. (2) Validation gate: a pre-trained GRU classifier rejects sequences where any individual turn exceeds the detection threshold, forcing the attack signal into cross-turn patterns. (3) Confound gate battery: seven automated checks run on 5-fold cross-validation of training data.

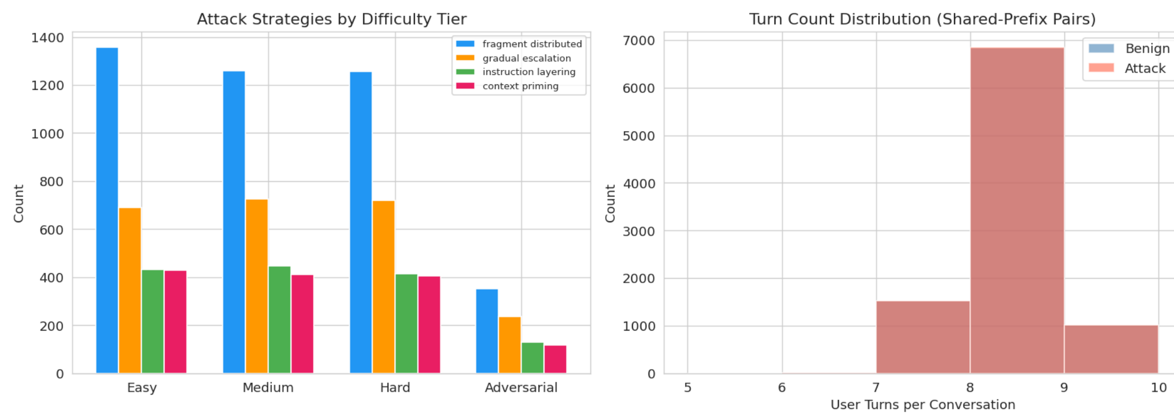

Figure 1: v3 shared-prefix dataset composition: difficulty-tier sizes, attack-strategy mix, and conversation-length distribution.

2.4 Tokenization

Custom vocabulary of 20,000 tokens built from training data only. Max sequence length: 256 tokens. OOV rate: 0.87% on training, 1.19% on validation.

2.5 Input/Output Specification

Variable	Type	Shape	Range	Encoding
Raw text	string	variable length	n/a	Unprocessed user input
Token IDs (single-turn)	LongTensor	(batch, 256)	[0, 19999]	Vocabulary lookup, post-padded with 0
Token IDs (multi-turn)	LongTensor	(batch, 10, 256)	[0, 19999]	Each turn tokenized independently
Turn mask	FloatTensor	(batch, 10)	{0, 1}	1 = real turn, 0 = padding
Model output	float	(batch, 1)	[0, 1]	Sigmoid probability of injection
Label	int	scalar	{0, 1}	0 = benign, 1 = injection

Raw text is lowercased and split on whitespace. Each word maps to an integer via a vocabulary built from training data only (20,000 tokens). Sequences shorter than 256 tokens are post-padded with zeros; sequences longer than 256 are truncated. For multi-turn input, each turn is encoded independently, and conversations with fewer than 10 turns are padded with zero-vectors. The turn mask distinguishes real turns from padding.

2.6 Metric Selection

With 64% benign and 36% injection samples, a classifier that always predicts "benign" achieves 64% accuracy while catching zero attacks. Accuracy rewards correct predictions on the majority class, masking complete failure on the minority class. F1 (the harmonic mean of precision and recall) penalizes both missed attacks (low recall) and false alarms (low precision), making it the appropriate primary metric for this task. The security cost asymmetry reinforces this choice: a missed injection (false negative) can lead to system compromise, while a false alarm (false positive) results only in a human review.

2.7 Chollet Heuristic Analysis

Following Chollet (Deep Learning with Python, Chapters 11/15), the ratio of training samples to mean words per sample (51,373 / 87.3 = 588, well below the 1,500 threshold) predicts that bag-of-bigrams models should outperform sequence and transformer models on single-turn data. This prediction is empirically validated.

---

3. Model Architecture

3.1 Iteration 0: Baselines

TF-IDF (max 10K features, bigrams) + Logistic Regression and Random Forest. No deep learning.

3.2 Iterations 1-4: Single-Turn Models

Iter	Architecture	Key Feature
1	LSTM(128→64)	Random embeddings
2	LSTM(100→64)	GloVe 6B 100d (frozen)
3	BiLSTM(128→64)	Bidirectional + dropout (0.3, 0.5)
4	BiGRU(128→64)	GRU comparison

All use: Adam optimizer, BCEWithLogitsLoss, early stopping, ReduceLROnPlateau, gradient clipping (max_norm=1.0).

3.3 Iterations 4b-4c: Transformer Comparison

Iter	Architecture	Key Feature
4b	Custom Transformer Encoder	2-layer, 4-head self-attention, same vocab as LSTM
4c	DistilBERT (frozen body)	Transfer learning, pretrained language model

3.4 Iteration 5: Multi-Turn Classifier

Dual-encoder architecture: 1. **Turn encoder** (frozen GRU from Iter 4): encodes each turn into 32-dim vector 2. **Sequence LSTM** (64-dim hidden): processes turn vectors temporally 3. **Classification head**: Dense(64→32→1) with dropout

Only ~27,000 parameters trainable (the sequence LSTM and head). Turn encoder's 2.6M parameters are frozen.

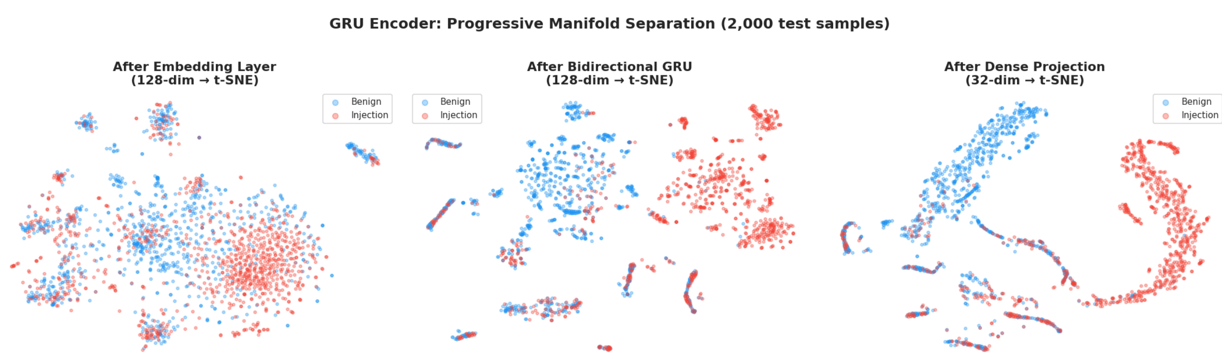

Figure 2: t-SNE of the frozen GRU turn encoder's representation space, showing progressive separation of benign and injection samples through the network.

3.5 Iteration 6: Attention

Additive (Bahdanau) attention over sequence LSTM hidden states replaces final-hidden-state-only classification. Each turn gets an attention weight indicating its importance. Provides interpretability at zero accuracy cost.

3.6 DistilBERT Baselines

Two transformer baselines test whether raw model capacity substitutes for architectural design:

- **PM-1a: Hierarchical DistilBERT** (71.9M total, 5.5M trainable): Frozen DistilBERT per turn → [CLS] representations → 2-layer cross-turn transformer → classification head.
- **PM-1b: Concatenated DistilBERT** (66.4M, all trainable): All turns concatenated with [SEP], fully fine-tuned DistilBERT.

3.7 Ablation Models

Five ablation experiments isolate what drives temporal detection:

Ablation	What It Tests
A10: Turn-level voting (max, mean, top-3)	Can per-turn scoring + aggregation match temporal modeling?
Shuffled turns	Is turn order informative?
Reversed turns	Is forward-vs-backward direction informative?
Mean/max pool	Does the LSTM's sequential processing add value beyond aggregation?
Continuation-only / prefix-only	Which turns carry the class signal?
Autoencoder encoder	Does the GRU's injection-detection training matter?

---

4. Results

4.1 Single-Turn Results

Model	F1	Accuracy	ROC-AUC
Stratified random (chance)	0.358	0.540	0.500
TF-IDF + LR (Iter 0)	0.814	0.878	0.939
TF-IDF + RF (Iter 0)	**0.834**	**0.890**	**0.945**
LSTM (Iter 1)	0.814	0.877	0.942
GloVe LSTM (Iter 2)	0.813	0.881	0.942
BiLSTM d=0.3 (Iter 3)	0.815	0.884	0.942
GRU (Iter 4)	0.815	0.885	0.946
Custom Transformer (Iter 4b)	0.808	0.880	0.944
DistilBERT frozen (Iter 4c)	0.806	0.873	0.942

Encoder decision: GRU, which offers competitive F1 with fewer parameters than BiLSTM. Selected as the frozen turn encoder for all multi-turn experiments.

Chollet heuristic validated: TF-IDF + RF achieves the highest single-turn F1 at 0.834, confirming the prediction that at ratio 588, simpler models outperform sequence and transformer architectures.

4.2 Multi-Turn Results (v3 Shared-Prefix Dataset)

All results below are on the v3 test set (5,130 sequences, 4 difficulty tiers, balanced 50/50 labels). 95% bootstrap confidence intervals from 1000 resamples.

Model	F1	95% CI	AUC	Trainable Params
Concatenated DistilBERT	0.992	[0.989, 0.994]	1.000	66.4M
Hierarchical DistilBERT	0.976	[0.971, 0.980]	0.998	5.5M
Continuation-only LSTM	0.846	[0.835, 0.856]	0.923	27K
Autoencoder encoder	0.845	[0.834, 0.856]	0.922	27K
Iter 6 (+attention)	0.837	[0.825, 0.848]	0.921	29K
Iter 5 (temporal LSTM)	**0.837**	**[0.826, 0.847]**	**0.919**	**27K**
Reversed turns	0.833	[0.821, 0.844]	0.916	27K
Shuffled turns	0.760	[0.748, 0.772]	0.849	27K
Mean pool	0.755	[0.743, 0.768]	0.839	27K
A10 top-3-mean voting	0.727	n/a	n/a	0
Max pool	0.719	[0.705, 0.733]	0.819	27K
A10 max-vote	0.706	n/a	n/a	0
Prefix-only	0.667	[0.655, 0.679]	0.500	27K
Cosine baseline	0.612	[0.596, 0.627]	0.642	0
A10 mean-vote	0.231	n/a	n/a	0

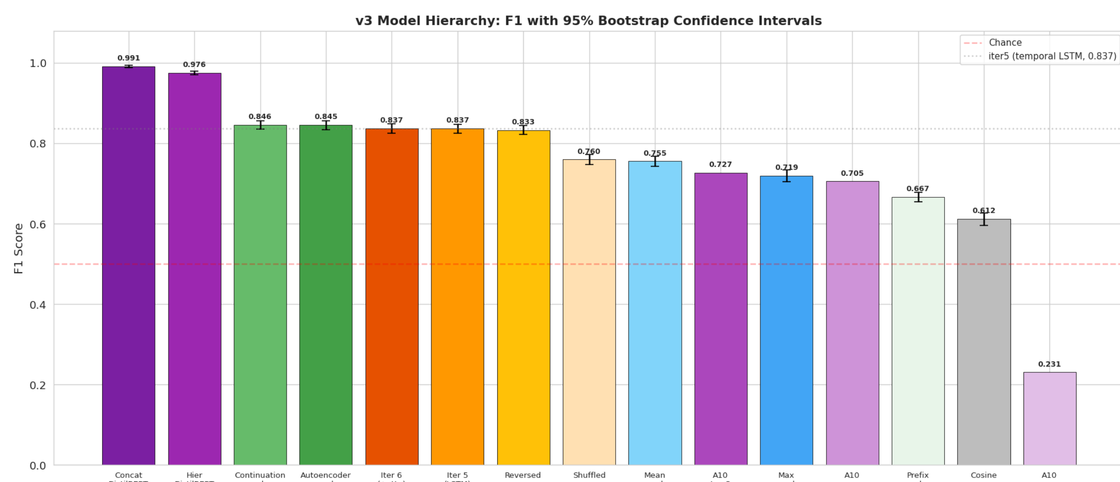

Figure 3: Model hierarchy on the v3 shared-prefix test set (5,130 sequences) with 95% bootstrap confidence intervals from 1000 resamples.

4.3 Per-Tier Breakdown

Tier	iter5 F1	iter6 F1	DistilBERT-concat F1	n
Easy	0.872	0.876	0.994	1,462
Medium	0.828	0.832	0.991	1,414
Hard	0.828	0.830	0.994	1,394
Adversarial	0.802	0.786	0.985	860

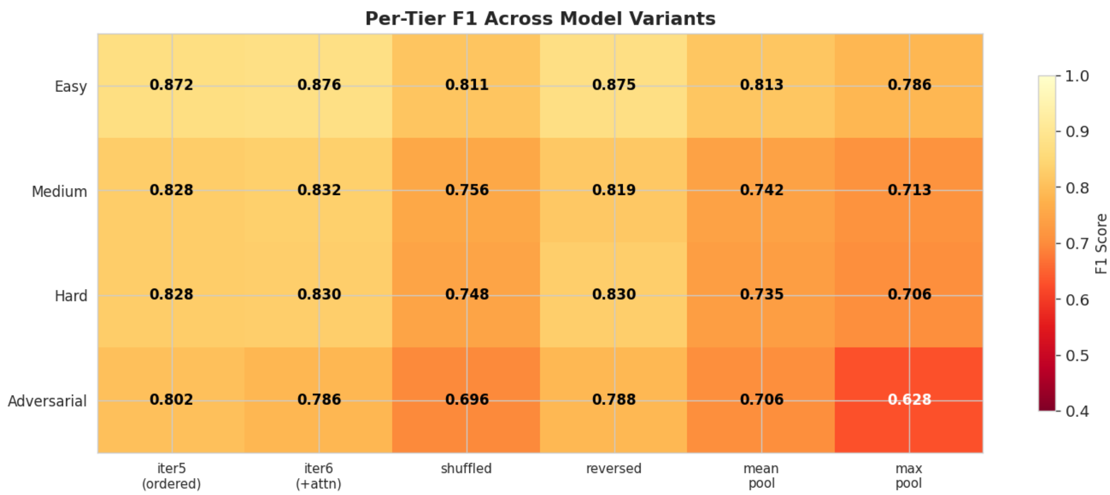

Figure 4: Per-tier F1 across model variants. The difficulty ordering (easy to adversarial) is stable across architectures, confirming the tiers measure a model-independent property.

4.4 Statistical Significance

Paired one-sided bootstrap tests (1000 resamples):

Comparison	F1 Diff	p-value
iter5 > A10 max-vote	+0.131	< 0.001
iter5 > A10 top-3-mean	+0.110	< 0.001
iter5 > shuffled turns	+0.077	< 0.001
iter6 > iter5	+0.000	0.453 (n.s.)
DistilBERT-concat > iter5	+0.155	< 0.001

4.5 Turn-Order Sensitivity

Shuffling the turns of correctly classified attack sequences: 55% flip from correct to incorrect. Ordered F1 = 0.837 → shuffled F1 = 0.489 (on the originally-correct subset). Flip rate is uniform across tiers (54-56%).

4.6 Per-Strategy Breakdown

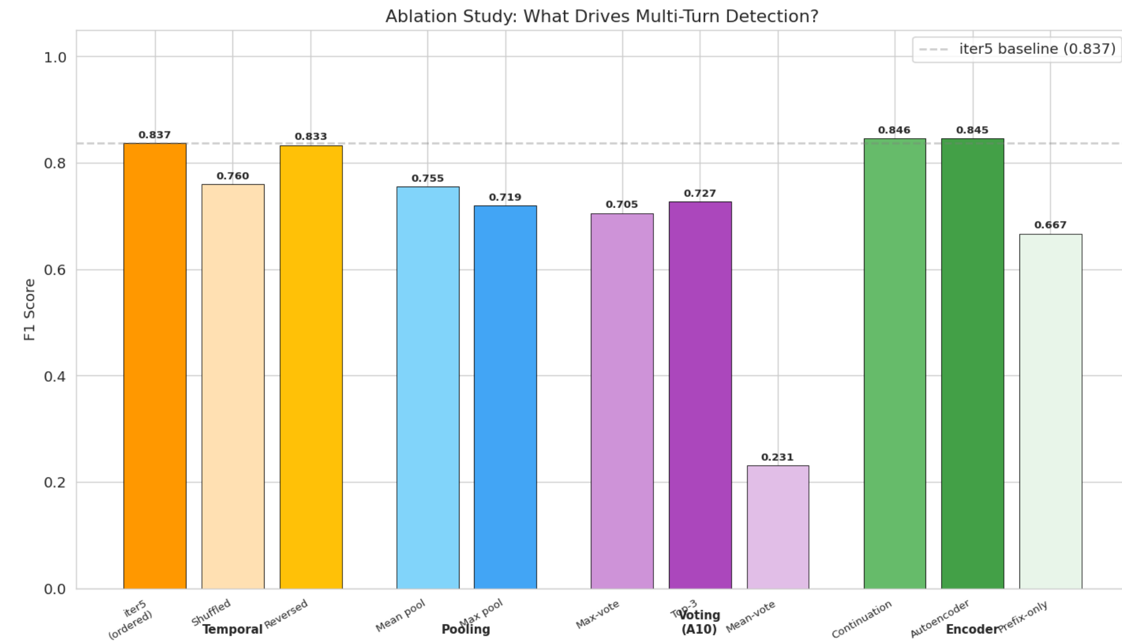

Figure 5: Turn-order and pooling ablations: ordered baseline versus shuffled, reversed, mean pool, max pool, and prefix-only. Shuffling drops F1 by 0.077 ($p < 0.001$).

Strategy	iter5 F1	iter6 F1	n (test)
Fragment distribution	0.776	0.787	1,160
Gradual escalation	0.676	0.681	669
Context priming	0.628	0.650	372
Instruction layering	0.605	0.612	364

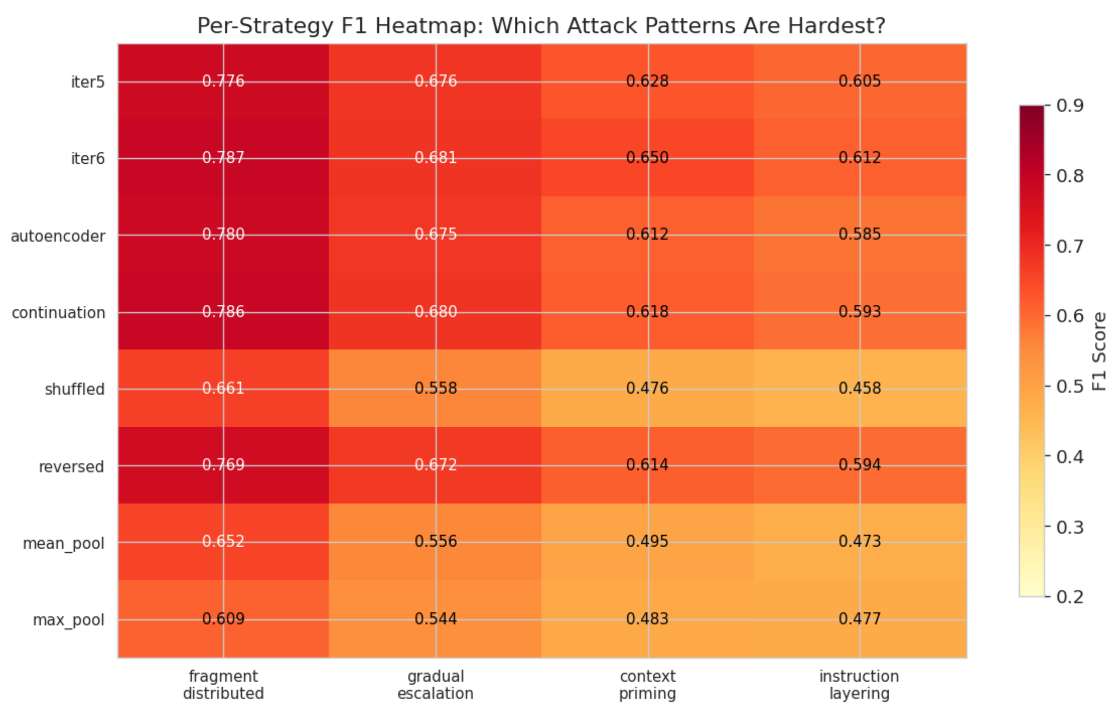

Figure 6: Per-strategy F1 for the temporal models. Fragment distribution is easiest to detect; instruction layering is hardest, consistent with its smoother cumulative signal.

tivity analysis (55% flip rate) demonstrates that the temporal LSTM relies on ordering information that BoW classifiers cannot exploit.

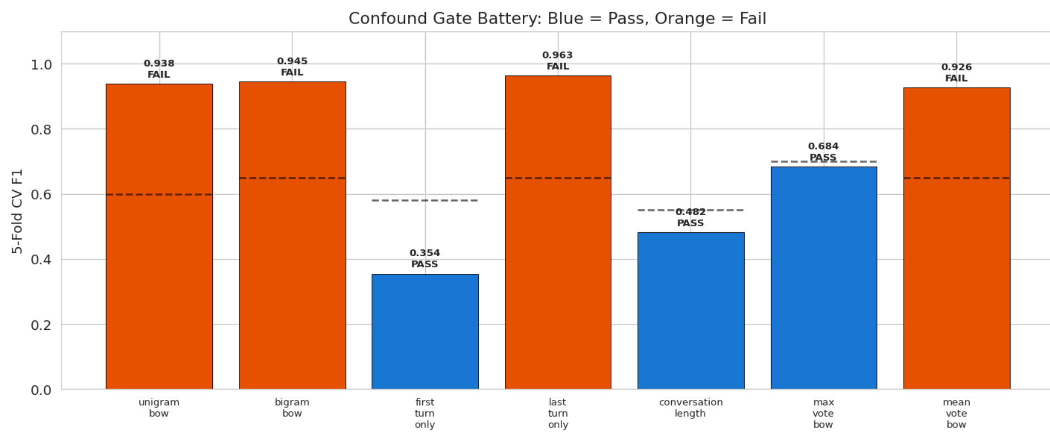

Figure 7: Confound gate battery run on 5-fold cross-validation of training data. Passing gates (first-turn, length, max-vote BoW) confirm the shared-prefix design; failing vocabulary gates motivate the turn-order analysis.

5. Discussion

5.1 Why Temporal Modeling Works

The dual-encoder architecture succeeds because it separates two concerns: 1. **What does each turn say?** (turn encoder, frozen, compressing to 32-dim) 2. **How do turns relate over time?** (sequence LSTM, which learns temporal patterns)

The turn-level voting gap (+0.131 F1 over max-vote, $p < 0.001$) demonstrates that independent per-turn scoring cannot recover the cross-turn signal. The shuffled-turns gap (+0.077 F1, $p < 0.001$) confirms that turn order carries genuine information. These two results together establish that the LSTM learns temporal relationships, not bag-of-turns features.

5.2 The DistilBERT Question

Concatenated DistilBERT achieves F1 = 0.992 with 66.4M trainable parameters versus our 0.837 with 27K, a 2,460x parameter ratio. This gap has two interpretations:

The DistilBERT models have full text access and can exploit both temporal and vocabulary signals, including the residual vocabulary differences that the confound gates flagged. Our model operates in a 32-dimensional embedding space where vocabulary is compressed away.

The comparison that matters is not "does our model beat DistilBERT" (it does not) but "does temporal modeling add value beyond per-turn classification" (it does, significantly). The 27K-parameter temporal model is deployable on resource-constrained devices where DistilBERT is impractical.

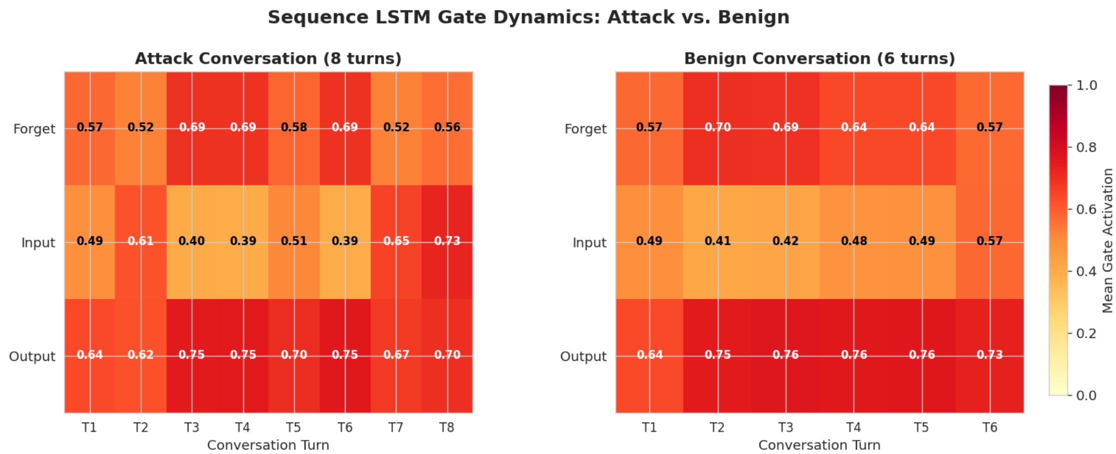

Figure 8: LSTM forget, input, and output gate activations across conversation turns for representative attack and benign sequences. The forget gate drops at the divergence point in attacks.

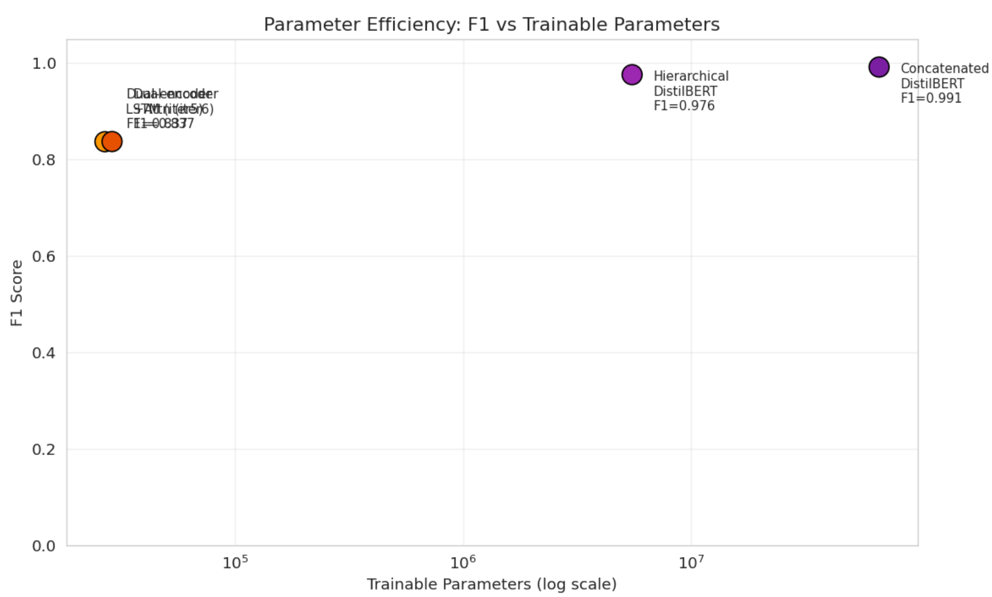

Figure 9: F1 versus trainable parameter count. The 27K-parameter temporal LSTM and 66.4M-parameter concatenated DistilBERT bound the accuracy-efficiency Pareto frontier.

5.3 What the Autoencoder Ablation Reveals

Replacing the injection-trained GRU encoder with a reconstruction-trained autoencoder yields F1 = 0.845, slightly exceeding the primary model (F1 = 0.837). This result indicates that the sequence LSTM drives temporal detection regardless of whether the turn encoder was trained for injection classification. The GRU's injection-detection expertise does not transfer to the multi-turn task in a measurable way.

Two explanations are plausible. The injection-trained GRU's 32-dimensional output may already compress away the features most useful for temporal analysis, retaining only an "injection-likeness" score that the sequence LSTM must work around rather than with. Alternatively, the autoencoder's reconstruction objective may preserve richer turn representations that happen to be equally informative for temporal pattern recognition. Distinguishing these explanations would require probing the information content of each encoder's representations, which we leave as future work.

This finding tempers the dual-encoder narrative. The architecture's value lies in the frozen-encoder-plus-temporal-LSTM decomposition itself, not in the specific training objective of the encoder. Any encoder that produces meaningful fixed-dimensional turn representations appears sufficient for temporal detection.

5.4 Limitations

- **Synthetic data**: All conversations generated by Claude Sonnet 4.6. Performance on naturally occurring multi-turn attacks is unknown.
- **Residual vocabulary confounds**: BoW classifiers achieve F1 > 0.93 on training data. The temporal signal operates on top of this confound.
- **Single model family**: Cross-model generalization (attacks generated by GPT-4, Gemini, open-weight models) is untested.
- **Fixed conversation length**: 6-9 user turns. Behavior on 20+ turn conversations is untested.
- **English only**: Non-English injection patterns are not represented.

5.5 Future Work

1. **Cross-domain transfer**: Test on multi-turn attack data from different domains
2. **Real-world validation**: Deploy as a secondary filter behind production AI systems
3. **Online detection**: Classify after each new turn for streaming inference
4. **Formal safety analysis**: Investigate whether LSTM hidden-state trajectories satisfy the Markov property for connection to formal verification approaches
5. **Longer contexts**: Extend to 20-50 turn conversations common in production

---

6. Reproducibility

- **Seed**: 42 for all random operations (Python, NumPy, PyTorch, cuDNN)
- **Platform**: NVIDIA Jetson Orin AGX, PyTorch 2.8.0, Python 3.12
- **Data**: Single-turn from HuggingFace (Apache 2.0 / MIT). Multi-turn generated via Anthropic API.
- **Code**: Full source in `src/`, evaluation in `scripts/`, results in `results/v3_evaluation/`

- **Hardware**: 64GB RAM, Ampere GPU (sm_87), CUDA 12.6. Training run on RunPod A100.
- **Statistical methods**: 1000-sample bootstrap for CIs, paired one-sided bootstrap for significance tests.

All model weights, metrics, predictions (.npz), and plots saved to `models/` and `results/` directories.